

Introduction to Services

Lab01

What is Service-Oriented Architecture?

- Service-Oriented Architecture (SOA) is an architectural style.
- Applications built using an SOA style deliver functionality as *services that can be used or reused when building applications or integrating within the enterprise or trading partners.*

SOA



Uses open standards to integrate software assets as services



Standardizes interactions of services



Services become building blocks that form business flows



Services can be reused by other applications

What is a Service?

A service is a reusable component that can be used as a building block to form larger, more complex business-application functionality.

A service may be as simple as “get me some person data,” or as complex as “process a payment.”

What is a Service?

- A service provides a discrete business function that operates on data.
- Its job is to ensure that the business functionality is applied consistently, returns predictable results, and operates within the quality of service required.

What is a Service?

- How the service is implemented, and how a user of the service accesses it, are limited only by the SOA infrastructure choices of the enterprise.
- From a theory point of view, it really doesn’t matter how a service is implemented.

Characteristics of a Service

- **<u>Supports open standards for integration</u>**: Although proprietary integration mechanisms may be offered by the SOA infrastructure, SOA's should be based on open standards.
- Open standards ensure the broadest integration compatibility opportunities.

Characteristics of a Service

- **Loose coupling**: The consumer of the service is required to provide only the stated data on the interface definition, and to expect only the specified results on the interface definition.



Characteristics of a Service

- **Stateless**: The service does not maintain state between invocations.
- It takes the parameters provided, performs the defined function, and returns the expected result.
- If a transaction is involved, the transaction is committed and the data is saved to the database.

Characteristics of a Service

- **Location agnostic**: Users of the service do not need to worry about the implementation details for accessing the service.
- The SOA infrastructure will provide standardized access mechanisms with service-level agreements.

SOA Benefits

- The services model simulates business needs better than objects or resources
  - IT Services (virus detection and elimination service)
  - Marketing services (promotional campaigns, brand marketing programs etc.)
  - Accounting services (Audits, Quarterly statements etc.)

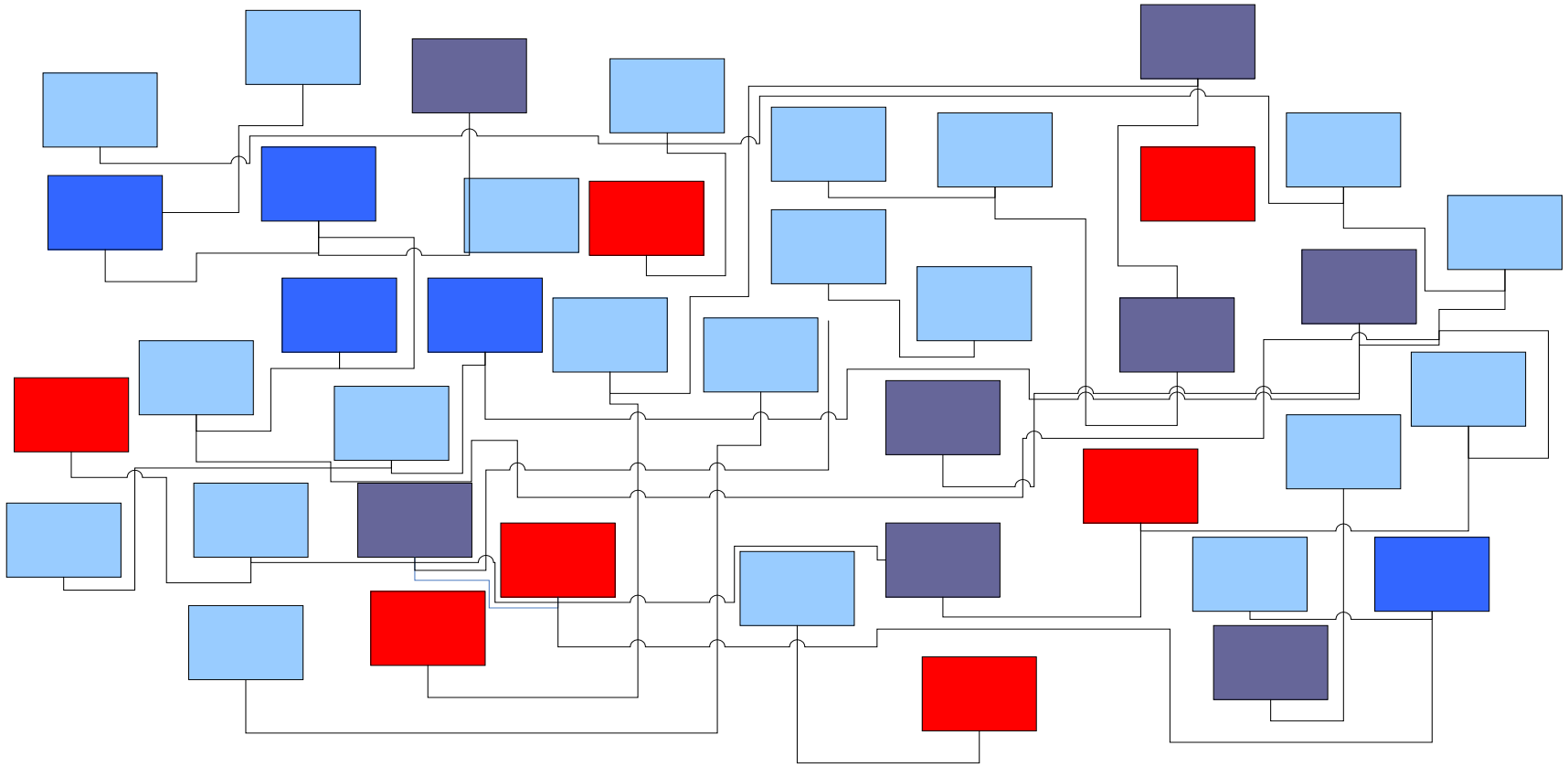
- Loose coupling between consumer and provider
  - Allows us to improve availability by having redundant providers.

- Dynamic binding allows the use of the latest services that are most useful to the task at hand.
  - In a highly adhoc networked environment, “discovery” becomes critical.
- Allows you to leverage existing IT assets – is a evolutionary architecture not a revolutionary one.

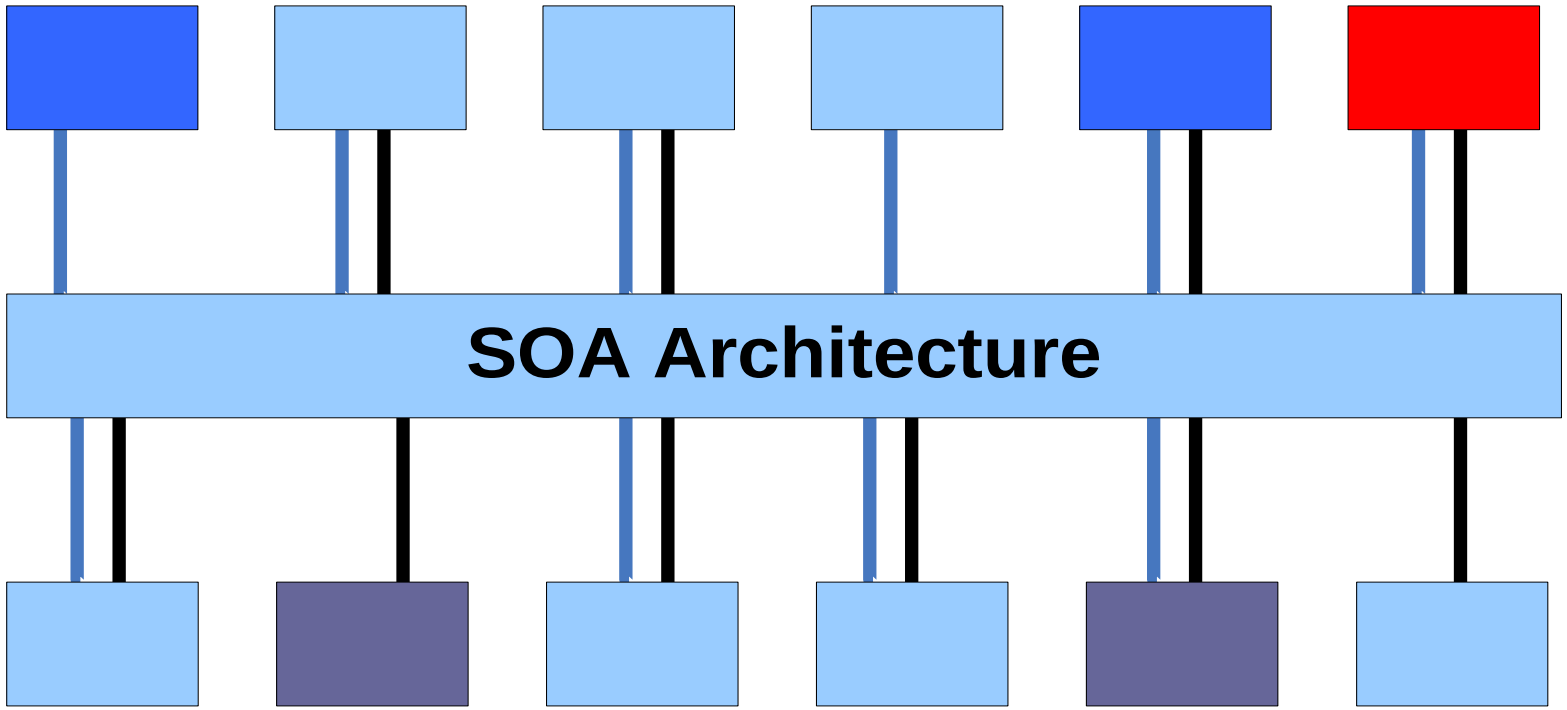
SOA Benefits - 2

- More cost efficient from an IT perspective:
  - Computing as a utility rather than as infrastructure
  - Use as you need model
  - Reuse
- Flexibility
  - Compose services on the fly

Legacy Integration



SOA Integration



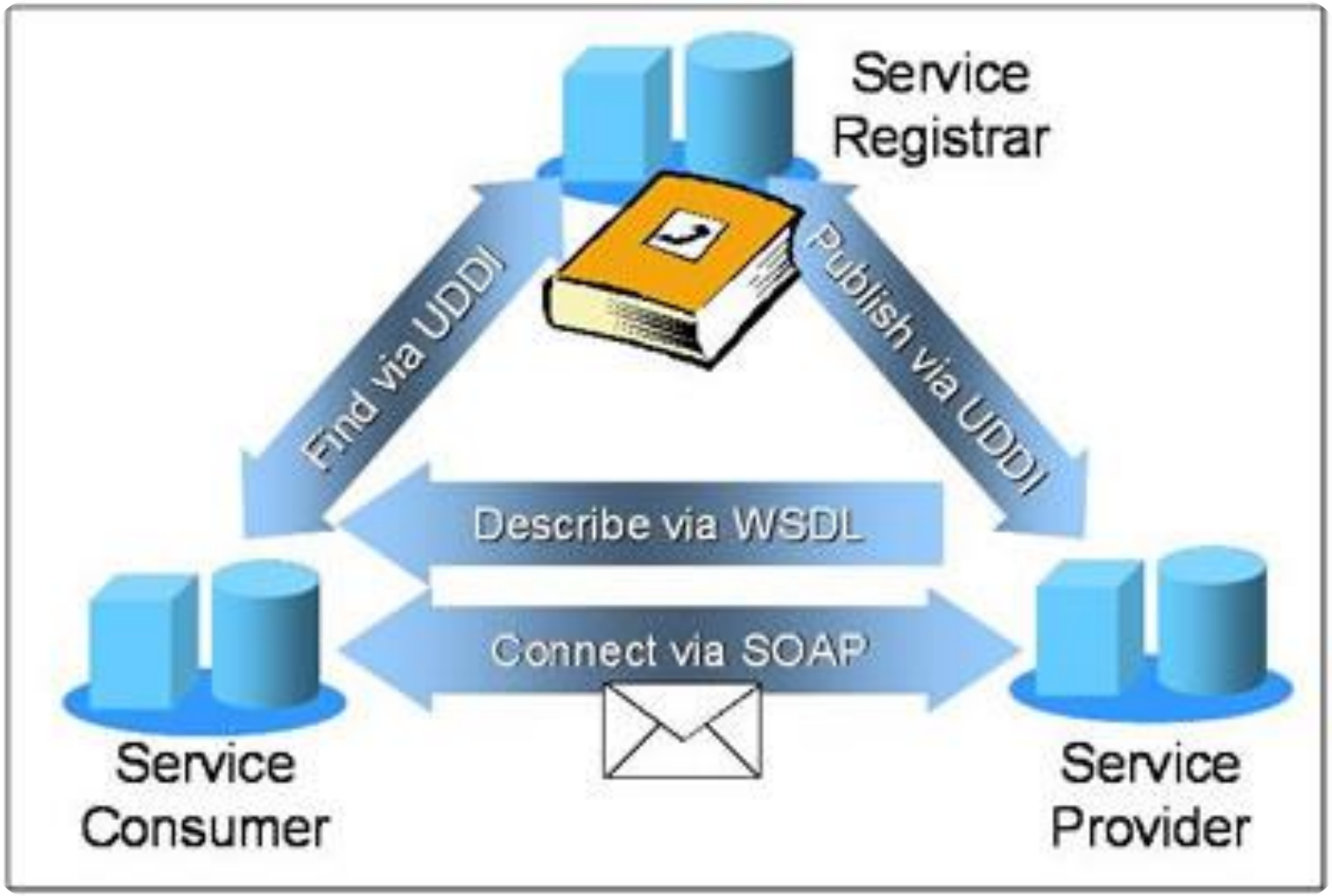
What is a Web service?

- Application capability, function or operation exposed to other programs via open, interoperable standards
  - “payloads” defined as XML documents
  - “transports” over http or other open Internet protocol
- Can be access from any programming language, hardware platform or operating system

What is a “Good Web Service”?

- “self describing”
  - XML Schema or DTD for grammar of payload
  - human readable instances
- discoverable
  - UDDI, WS-Inspection, or some other method of advertising availability
  - Realize promise of many to many integrations that are enabled by a web service (one to one ad hoc integrations can be done with earlier technology)
- “coarse grained”
  - Large documents representing entire business events or business objects
  - e.g. CRUD operations: CreateOrder, ReadOrder, UpdateOrder, DeleteOrder
  - NOT operations such as SetPrice(), SetProduct(), SetCustomer(), CompleteOrder()
- asynchronous
  - When appropriate (often for large scale work)

Base Protocols in Web Services



Standards of Services

- The three main standards that enable implementation of Web services are
- The Simple Object Access Protocol (SOAP),
- Web Services Description Language (WSDL), and
- Universal Description Discovery, and Integration (UDDI).



SOAP

- SOAP is an XML-based protocol to support communication between a Web service,
- its clients, and UDDI registry.
- It facilitates communication between Web services and interacting application components across heterogeneous platforms without requiring the use of custom binaries or other platform specific information .
- It provides a mechanism to invoke a Web service.

WSDL

- WSDL is an XML-based standardized interface definition language used to describe what a Web service can do, where it resides, and how it can be invoked.
- A WSDL file associated with a Web service contains important details about the Web-service interface for client-service interaction.
- WSDL is used to provide definition of a Web service and interface specification for it.

UDDI

- UDDI standard is used to publish, discover, and manage Web services in an UDDI registry.
- UDDI registry is like Yellow Pages that list available Web services that can be discovered by an application.
- Creation of a registry that contains available services allows service consumers to discover and invoke Web services that are published within the UDDI registry.